Ledger saturation: arithmetic closes, phase diamonds remain

Draft (2026-08-10). Research report for the `ledger-saturation` track. This report does not change the observer language of `local-classes.md`.

Claim labels used throughout:

- **Verified** means checked against an accepted AIP, a mechanized repository fact, or the exact finite audit described here.
- **Desk-proved** means a complete mathematical argument is given but has not been mechanized.
- **Conjectured** means a precise statement is proposed without a proof.
- **Wall** means an implication fails or stops at the named missing lemma.

1. Verdict first

Desk-proved, correction. The integral Noetherian wall in `cocycle-classification.md` §5.7 is not a wall for the relation lattices that classify tables. For every observer and grade,

$$\mathcal{R}_n = \left\{(t, \lambda) \in \mathbb{Z}^D \oplus \mathbb{Z}^\Sigma \mid \rho_n(t) - \ell_n^* \lambda \text{ is an integral gradient}\right\}$$

is saturated in the fixed lattice $\mathbb{Z}^{D+\Sigma}$:

$$\mathcal{R}_n = \left\{x \in \mathbb{Z}^{D+\Sigma} : x \in \operatorname{span}_{\mathbb{Q}}(\mathcal{R}_n)\right\}.$$

Consequently two relation lattices are equal exactly when their rational spans are equal. The descending chain $\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset \cdots$ cannot occur here; its proper terms are not saturated. Fixed-grade arithmetic torsion is still real, but it appears only after projecting away the price coordinates.

Desk-proved, route (b). Let $M(A)$ be the finite transformation monoid generated by F, W, L, P, and let $e(A)$ be the least common multiple of the exponents of its maximal subgroups. At every grade, if $B_n \leq \mathbb{Z}^\Sigma$ is the signed letter-count lattice of reachable augmented cycles, then

$$e(A)H \subseteq B_n \subseteq H, \qquad H = \left\{x : x_{up} = x_{down0} + x_{down1}\right\}.$$

Thus

$$\operatorname{Tor}\,(\mathbb{Z}^\Sigma/B_n) = H/B_n \quad \text{and} \quad e(A)\,\operatorname{Tor}\,(\mathbb{Z}^\Sigma/B_n) = 0.$$

This is the proposed maximal-subgroup theorem for **all** finite observers, not only permutation observers. If A is aperiodic, $e(A) = 1$, so $B_n = H$ and there is no arithmetic torsion at any grade. Resets add no torsion. All arithmetic periods come from the group part.

Desk-proved, named observers complete. There are two independent descent conditions:

1. if every reachable augmented graph is strongly connected, the round-2 periodic-closure argument gives $\mathcal{R}_{n+1} \subseteq \mathcal{R}_n$;
2. if $W = \mathrm{id}$, the open left-subtree replay is an exact graph lift, so the same inclusion holds without strong connectivity.

In either case the rational spans form a descending chain in a fixed finite dimensional space and stabilize; (1) upgrades this immediately to integral stabilization. Dirty satisfies the first condition. First touch and warmth satisfy the second. Their monoids are aperiodic, so (3) also proves that their fixed-grade arithmetic parts are torsion-free. The whole full-table tower theory therefore has eventually stable integral relation lattices for every named irreversible observer. The already known named-cost thresholds remain sharper: first touch is trivial at every grade, dirty is nontrivial at every positive grade, and warmth is nontrivial from grade two.

Wall, reduced to the free part. For a general observer with both $W \neq \mathrm{id}$ and non-strongly-connected reachable graphs, the remaining issue is not an exponent. It is rational realization of signed path differences between strongly connected components. Contracting the components separates the problem into recurrent cycles, which (2) controls, and a free **phase-diamond lattice**. An aperiodic reset can destroy the lift of a phase diamond completely, producing a free cokernel even though every maximal subgroup is trivial. A four-vertex abstract reset diamond below is the minimal obstruction to any monoid-only proof.

The exact machine-specific lemma still needed is:

> **Dyadic reset-diamond filling.** In the confined left-subtree lift of every reachable $G_{A,n}$, the projection of reachable signed cycles contains the phase-diamond lattice after tensoring with $\mathbb{Q}$.

No proof or dyadic counterexample is supplied. This is strictly smaller than the previous "uniform integral ledger saturation" wall: saturation and all arithmetic indices are now settled; only a rational connector theorem for irreversible phases remains.

2. The object: an address-resolved reachable event ledger

The phrase "event ledger" in the round-2 report conflated three groups. This section separates the ambient event module, the compatible marginal lattice, and the actually realized signed-cycle lattice.

2.1 Reachable augmented graph

Fix a grade n, one head, and an observer $A = (Q, q_0, F, W, L, P)$. Multiple addressed heads only add the head index to the action slots below. Write

$$N_n = \{0,1\}^{\leq n}, \qquad \Lambda_n = \{0,1\}^n$$

for the nodes and leaves of the complete tree. Let $G_{A,n} = (V_n, E_n)$ be the directed multigraph defined in `local-classes.md` §2: its initial set has arbitrary legal base memory and head position, all marks equal to q_0, and its vertices are the forward closure of that set. Parallel runnable actions remain distinct edges. Put

$$Z_1(G_{A,n}) = \ker\left(\ \partial_G : \mathbb{Z}^{E_n} o \mathbb{Z}^{V_n}\ \right) = H_1(G_{A,n}; \mathbb{Z}).$$

These are signed cycles in the underlying multigraph. They are not merely directed closed walks.

2.2 Event slots

The free event module C_n^{evt} has the following basis. A slot is retained only if it occurs on some edge of the reachable graph.

- $d(v, q)$ for a down into a non-root node v whose pre-fill mark is q. The last bit of v distinguishes `down0` from `down1`.
- $u(v, q)$ for an up from v whose pre-leave mark is q.
- $r(x, q, m)$ for a read at leaf x with mark q and memory bit m.
- $z_b(x, q, m)$ for `writeb` at leaf x with pre-write mark q and memory bit m.
- $p(v, q)$ for the collateral P update at a node of depth at least two.
- $w(v, q)$ for the collateral W update at a non-root internal node.

The leaf write slot z_b already records the leaf's W update; w records only strict non-root ancestors. Collateral slots are unpriced. They are essential: erasing them loses the P/F correlation detected by warmth and the W/L correlation detected by dirty write-back.

Every reachable global edge e has an event vector

$$\epsilon_n(e) \in C_n^{\mathrm{evt}}.$$

A down into v contributes $d(v, q_v)$ and one $p(vb, q_{vb})$ for each child vb of v. An up contributes $u(v, q_v)$. A read contributes $r(x, q_x, m_x)$. A write at x contributes $z_b(x, q_x, m_x)$ and $w(v, q_v)$ at every strict non-root ancestor v of x. Extend (5) linearly to $\mathbb{Z}^{E_n}$.

2.3 The compatible marginal lattice

There are three incidence maps on C_n^{evt}.

1. The observer boundary at node v sends an event with update $U \in \{F, L, W, P\}$ and pre-state q to $[Uq] - [q]$ in $\mathbb{Z}^Q$. Reads have zero observer boundary.
2. The memory boundary at leaf x sends $z_b(x, q, m)$ to $[b] - [m]$ in $\mathbb{Z}^{\{0,1\}}$; reads are loops.
3. The head boundary sends a down into v to $[v] - [\mathrm{parent}\ (v)]$ and an up from v to its negative.

The synchronization map records

$$\sum_q p(v, q) = \sum_q d(\mathrm{parent}\ (v), q) \qquad (\mathrm{depth}\ (v) \geq 2),$$
$$\sum_q w(v, q) = \sum_{x \in \Lambda_n,\, x \succ v} \sum_{b,q,m} z_b(x, q, m) (0 < \mathrm{depth}\ (v) < n).$$

Here an equality means that the difference is a row of the synchronization map. Define the lattice of compatible address-resolved marginals by

$$\mathrm{Comp}_n(A) = \ker(\partial_Q \oplus \partial_{\mathrm{mem}} \oplus \partial_{\mathrm{head}} \oplus s_n) \leq C_n^{\mathrm{evt}}.$$

Because it is the kernel of a homomorphism between free abelian groups, $\mathrm{Comp}_n(A)$ is saturated. Equations (6) are aggregate constraints; they deliberately do not assert that arbitrary compatible marginals arise from one global execution.

2.4 The reachable signed-cycle event ledger

Definition. The grade-n *reachable signed-cycle event ledger* is

$$\mathrm{Led}_n(A) = \epsilon_n\ \big(\ Z_1(G_{A,n})\ \big) \leq \mathrm{Comp}_n(A).$$

This is the requested object. It is an image lattice, not the kernel (7). Its realization defect is

$$\mathrm{Def}_n(A) = \mathrm{Comp}_n(A)/\mathrm{Led}_n(A).$$

The defect can have a free part. Its finite saturation defect is separately

$$\mathrm{SatDef}_n(A) = \frac{\{x \in \mathrm{Comp}_n(A) : x \in \mathrm{span}_{\mathbb{Q}}(\mathrm{Led}_n(A))\}}{\mathrm{Led}_n(A)}.$$

The round-2 statement that the cokernel has bounded exponent silently asserted both that (9) is finite and that (10) has a uniform exponent. Those are different assertions. The present result settles exponent questions for letter-count monodromy, not rational surjectivity of (8).

A table $T : \Sigma \times Q \to \mathbb{Z}$ is the functional on (7) that evaluates the four action-slot families using their raw operation and pre-state and vanishes on collateral slots. A static price λ ignores the pre-state. Thus both table holonomy and word-price holonomy factor through (8). The smaller compressed module $\mathbb{Z}^{\Sigma \times Q}$ is adequate for evaluating a fixed table but not for stating realization: it forgets the addressed gluing that creates warmth and dirty holonomy.

2.5 What the left-subtree embedding actually induces

There is no canonical homomorphism $\mathrm{Led}_n(A) \to \mathrm{Led}_{n+1}(A)$. Replaying a path inside the new left subtree leaves all old events unchanged but turns the old root into a non-root boundary node. A word with k writes changes its mark by

$$q \mapsto W^k q.$$

The correct functorial object is the confined lift graph $\hat{G}^{,0}_{A,n+1}$: the reachable portion of $G_{A,n+1}$ whose head never leaves the left subtree, together with the projection

$$\pi_n : \hat{G}^{,0}_{A,n+1} \to G_{A,n}$$

that forgets the new root, right subtree, and boundary mark. The induced map goes covariantly on realized cycles,

$$(\pi_n)_* : H_1(\hat{G}^{,0}_{A,n+1}; \mathbb{Z}) \to H_1(G_{A,n}; \mathbb{Z}).$$

Equivalently, one can retain relative ledgers indexed by the boundary pair $(q, W^k q)$. Treating open replay as a cycle inclusion is exactly the reachability mistake that the saturation conjecture was trying to repair.

3. Saturation of the full relation lattice

Let

$$a_n : \mathbb{Z}^{D+\Sigma} \to \mathbb{Z}^{E_n}, \qquad a_n(t, \lambda) = \rho_n(t) - \ell_n^* \lambda.$$

Let $\nabla_n = \delta \mathbb{Z}^{V_n}$ be the integral gradient lattice.

Lemma, desk-proved. ∇_n is saturated in $\mathbb{Z}^{E_n}$.

Proof. If $kc = \delta\Phi$, evaluation on every signed cycle gives $kc(z) = 0$ in $\mathbb{Z}$, hence $c(z) = 0$. The spanning-forest construction integrates the integer edge values of c to an integer vertex potential. Thus $c \in \nabla_n$. □

Theorem, desk-proved (automatic relation saturation). For every finite observer and every grade, $\mathcal{R}_n = a_n^{-1}(\nabla_n)$ is saturated and satisfies (1).

Proof. A preimage of a saturated subgroup is saturated. The rational points of the same homogeneous equations are precisely $\operatorname{span}_{\mathbb{Q}}(\mathcal{R}_n)$. □

Corollary, desk-proved. If the rational relation spaces are eventually constant, then the integral relation lattices are eventually constant at the same grade. If they form a descending chain, finite-dimensionality over $\mathbb{Q}$ supplies such a grade.

This does not erase visit-parity torsion. The projection

$$\{(t, \lambda) : (t, \lambda) \in \mathcal{R}_n\} \to \mathbb{Z}^D$$

need not have saturated image: a projection of a saturated lattice need not be saturated. For parity, twice the table pairs with an odd down-price vector; the entire pair is not twice an integral point of $\mathcal{R}_n$. The arithmetic class therefore survives exactly where the fixed-grade exact sequence placed it, without creating a second tower-level integral wall.

4. Route (b): maximal subgroups control all arithmetic torsion

4.1 A finite complete event automaton

First ignore the tree. Let a finite alphabet U act totally on a finite set Q, let $M \leq \operatorname{End}(Q)$ be its transformation monoid, and let $B \leq \mathbb{Z}^U$ be the signed letter-count image of the underlying event graph's cycle lattice.

Lemma, desk-proved (local monoid quotient). $\mathbb{Z}^U/B$ is a finite group quotient of M. Its exponent divides $e(M)$; if M is aperiodic, the quotient is trivial.

Proof. Choose a base state and assign to each state q the signed count of an undirected path from the base to q, modulo B. Cycle differences make this value $p(q)$ well-defined. For a letter u,

$$p(qu) = p(q) + [u].$$

The finite set $p(Q)$ is stable under translation by $[u]$. Translation in an abelian group is injective, hence it permutes this finite set. Because the base has value zero, the resulting translation action is faithful. If two words induce the same transformation of Q, (16) says that they induce the same translation. Thus the translation group factors through M and is exactly the group generated by the letter classes, namely $\mathbb{Z}^U/B$.

For any $m \in M$, the eventual period of the powers of m is the order of an element in a maximal subgroup. Hence $m^{N+e(M)} = m^N$ for sufficiently large N. In every group quotient this implies $m^{e(M)} = 1$. Aperiodic means every such period is one. □

As a finite stress test, all complete two-letter automata on two, three, and four states were enumerated: respectively $16, 729$, and $65, 536$ automata. Their cycle-count lattices were computed from fundamental cycles. No aperiodic monoid produced torsion, and every torsion exponent divided the least common multiple of the monoid's element periods. This audit is evidence for the implementation interface; the lemma does not depend on it.

4.2 The dyadic action graph

Let X_n be the unobserved grade-n action graph. It is strongly connected, and its signed letter-count lattice is the primitive rank-five lattice

$$H = \{x : x_{\mathit{up}} = x_{\mathit{down0}} + x_{\mathit{down1}}\}.$$

The down/up excursions of each orientation and the read, write-zero, and write-one loops generate (17).

Theorem, desk-proved (reset observers add no new arithmetic period). For every observer A and grade n, equations (2) and (3) hold.

Proof. Take a directed closed walk w in X_n based at x. The augmented initial vertex $(x, q_0^{N_n})$ belongs to the standard initial set. Repeating w acts on the vector of node marks by a coordinatewise transformation $\tau_w \in M(A)^{N_n}$: the base path fixes which F, W, L, P updates each coordinate receives, independently of its current mark.

The forward orbit of $q_0^{N_n}$ under τ_w is eventually periodic. Its period r divides $e(A)$, because every coordinate period does. From a reachable periodic point, w^r is a directed closed walk in $G_{A,n}$ with letter count $rN(w)$. Repeating it $e(A)/r$ times gives $e(A)N(w) \in B_n$. Directed closed walks generate $H_1(X_n; \mathbb{Z})$, so $e(A)H \subseteq B_n$. The reverse inclusion follows by forgetting marks. Because H is primitive, the torsion of $\mathbb{Z}^\Sigma / B_n$ is exactly H/B_n. If A is aperiodic, every $r = 1$ and $B_n = H$. □

This proof uses standard reachability rather than all mark assignments: the periodic point is reached by the transient copies of the same base loop. It also permits irreversible updates. Permutations were stronger than needed.

4.3 Named observers

First touch is generated by identity and the constant `seen` map. Dirty is generated by identity and the two constant maps `clean` and `dirty`. Warmth is generated by identity and the two constant maps `cold` and `hot`. Their transformation monoids are aperiodic. Therefore

$$B_n = H$$

for all three at every grade. The absence of torsion seen in the grade-one and grade-two Smith computations is a theorem, not a small-grade pattern.

The same statement holds for any synchronous product whose generated monoid is aperiodic. In particular, combining irreversible reset observers can create new free scheduling holonomy but cannot create an arithmetic class.

5. Route (a): SCCs isolate the remaining free module

5.1 Recurrent cycles versus phase diamonds

Let $\{C_i\}$ be the strongly connected components of $G_{A,n}$. Contract each C_i to a vertex but retain every intercomponent edge separately; call the resulting directed acyclic multigraph $D_n^\sharp$. Graph contraction gives a noncanonical split exact sequence

$$0 \to \bigoplus_i H_1(C_i;\mathbb{Z}) \to H_1(G_{A,n};\mathbb{Z}) \to H_1(|\ D_n^\sharp\ |;\mathbb{Z}) \to 0.$$

The first term is generated by directed closed walks. Periodic closure of the finite boundary orbit handles it, with periods bounded by (2). The last term is free and is generated by signed differences of directed paths through the component DAG with the same endpoints. Call it

$$\mathrm{Phase}_n(A) = H_1(|\ D_n^\sharp\ |;\mathbb{Z}).$$

This is the exact correction module omitted by a directed-loop test.

5.2 The component DAG does not stabilize

Wall, route (a) in its naive finite-correction form. The local event graph on Q has a fixed component DAG, but the global phase records one component per geometric node. For first touch, each newly touched node moves irreversibly from `untouched` to `seen`. Increasing the grade introduces new independent phase transitions and directed chains of strictly increasing length. The global condensation is a reachable subposet of a power of the two-element phase order; it cannot become a fixed finite DAG under the left-subtree embedding.

This does not obstruct first touch—its potential solves the problem—but it rules out the proposed proof "there are finitely many component transitions, so only finitely many corrections." A successful contraction theorem must use the tree equations and address symmetry to fill (20), not stabilization of the literal component DAG.

5.3 Exact descent slices

Theorem, desk-proved ($W = \mathrm{id}$ descent). If $W = \mathrm{id}$, then

$$\mathcal{R}_{n+1} \subseteq \mathcal{R}_n$$

for every n, with no connectivity hypothesis.

Proof. Replay a history reaching a grade-n vertex inside the left subtree. All old events and marks agree. The only extra update in the round-2 open lift is W on the old root, so its boundary mark remains q_0. This makes the replay independent of the chosen history and embeds the entire reachable underlying multigraph $G_{A,n}$ into the confined part of $G_{A,n+1}$, preserving raw labels and table entries. Restricting a grade-$(n+1)$ exchange equation to this subgraph gives (21). □

Corollary, desk-proved. Every observer with $W = \mathrm{id}$ has eventually stable integral relation lattices. This includes first touch, visit modulo r, and warmth, even though warmth is not strongly connected and has $P \neq \mathrm{id}$.

Theorem, desk-proved (strong reset stabilization). If every $G_{A,n}$ is strongly connected, its integral relation lattices eventually stabilize.

Proof. Round 2 proves (21) by periodic closure and proves that the rational chain stabilizes. Apply (1). □

This includes dirty and upgrades the previous "rational only" full-table statement to an integral theorem.

6. Route (c): the minimal obstruction to a monoid-only proof

Consider the directed diamond with two paths $s \to a \to t$ and $s \to b \to t$. Give a boundary bit initial value zero. The first edge on the upper path applies the constant reset $R(0) = R(1) = 1$; all other edges apply the identity. Include a zero-boundary source over every base vertex, matching the liberal standard source convention.

Verified, finite counterexample (abstract lift). The base underlying graph has $H_1 \cong \mathbb{Z}$, generated by the signed difference of the two paths. In the reachable lift, the upper path from $(s, 0)$ ends at $(t, 1)$ and the lower path ends at $(t, 0)$. Including the other zero-boundary sources adds the arm $(a, 0) \to (t, 0)$ but no connector to $(t, 1)$. The lifted underlying graph is a tree, so its H_1 is zero. The projection cokernel therefore has a free $\mathbb{Z}$ summand.

The transformation monoid $\{1, R\}$ is aperiodic and has only trivial maximal subgroups. Hence maximal-subgroup exponents control recurrent torsion but say nothing about rational lifting of phase diamonds. This is why (2) can be fully true while the realization map (8) is not rationally onto.

The diamond is not claimed to be realizable by the dyadic observer language. The machine supplies many additional paths, arbitrary initial memory and head positions, and rigid tree-coupling equations that may always create a connector. Establishing that fact is now the whole wall.

Conjectured (dyadic reset-diamond filling). Let $K_n = \mathrm{im}\ (\pi_n)_*$ in (13). After quotienting $H_1(G_{A,n})$ by its component-internal cycles, the image of K_n spans $\mathrm{Phase}_n(A)$ over $\mathbb{Q}$:

$$\mathrm{span}_{\mathbb{Q}}\ \Big(\ K_n + \sum_i H_1(C_i)\ \Big) = H_1(G_{A,n}; \mathbb{Q}).$$

An equivalent constructive form asks for every fundamental signed path difference in $D_n^{\sharp}$ to admit a nonzero integer multiple whose two open lifts can be connected in the standard-reachable boundary fiber. Equation (22), not a Smith exponent, is the next theorem or counterexample target.

If (22) holds, component-internal periodic closure and the phase spanning statement together imply $\mathcal{R}_{n+1} \subseteq \mathcal{R}_n$. The rational chain then stabilizes by dimension, and (1) makes the stabilized relation lattice integral. Thus (22) is sufficient for the remaining full tower theorem; it is not merely a diagnostic property of the ledger.

A smallest dyadic counterexample search should begin with an aperiodic synchronous product having:

- a warmth-like P reset, to make $G_{A,n}$ non-strongly-connected; and
- a dirty-like nontrivial W, to make the old-root boundary remember which arm of a phase diamond performed a write.

The individual named observers avoid this conjunction: warmth has $W = \mathrm{id}$, while dirty is strongly connected.

7. Consequence bookkeeping: the new dekkingskaart

slice	fixed-grade arithmetic	full relation tower	boundary
arbitrary finite observer	finite SNF; torsion killed by $e(A)$	open in general	rational phase-diamond filling (22)
aperiodic observer	no arithmetic torsion at any grade	open only if non-strong and $W \neq \mathrm{id}$	free scheduling/ phase part only
$W = \mathrm{id}$	as above	integrally eventually stable	complete
all $G_{A,n}$ strongly connected	as above	integrally eventually stable	complete
$W = P = \mathrm{id}$	finite one-leaf classifier	grade one decides all grades	complete, sharper than eventual stability
permutation observer	torsion killed by group exponent	integrally eventually stable	complete (round 2)
first touch	torsion-free	grade one; named class trivial	complete
dirty	torsion-free	integrally eventually stable; named class nonzero for all $n \geq 1$	complete
warmth	torsion-free	integrally eventually stable; named class nonzero for all $n \geq 2$	complete

“Integrally eventually stable” classifies the full formal-difference relation lattice. The nonnegative cone remains a separate Presburger intersection at each of the finitely many relevant grades, as in the earlier reports. Nothing here weakens the distinction between formal group completion and realizable natural prices.

The remaining general slice is narrower than “irreversible observers.” It requires irreversible phase structure and a nontrivial write-below boundary at the same time. Its arithmetic part is already complete, including mixed monoids: only the rational free part can fail.

8. Lean readiness and finite checks

The following statements are finite or generic graph algebra and can be mechanized without solving (22).

1. Define the reachable slot type of §2.2 and the edge ledger map ϵ_n; prove that a global cycle satisfies (6) and the three local boundary equations.
2. Define `Comp` as the kernel (7), `Led` as the image (8), and expose both the free rank and Smith invariants of (9). Do not call (9) an exponent before checking that its free rank is zero.
3. Prove integral gradients saturated by the existing spanning-tree cycle criterion, then derive (1) as a preimage lemma. This is the shortest new Lean theorem and removes the false Noetherian rung.
4. Represent a finite transformation monoid by its action on `Fin q`. Prove eventual-period divisibility by the maximal-subgroup exponent, then prove

   (2) by iterating a base loop from the standard mark vector.
5. For an aperiodic monoid use the checkable identity $m^N = m^{N+1}$ for a common finite N to specialize (2) to $B_n = H$.
6. Formalize the exact $W = \mathrm{id}$ left-subtree graph embedding and derive (21) without a directed-loop criterion.
7. Contract SCCs and identify the quotient (20). The abstract reset diamond is a six-vertex finite graph check suitable as a regression theorem.

At small grades, an external exact checker can enumerate $G_{A,n}$, construct the sparse integer matrix of ϵ_n, compute bases for (7) and (8), and report:

$$\operatorname{rank}\ \operatorname{Def}_n(A), \qquad \operatorname{SNF}\ \operatorname{SatDef}_n(A).$$

The first counterexample target is the warmth×dirty aperiodic product at the smallest grade where P reaches children. A positive free rank in (23) is a counterexample to (22); a finite cokernel is only evidence and must not be reported as a uniform theorem.

The genuinely infinitary task is proving that the family of phase modules (20), whose dimensions grow with the tree, has dyadic connectors uniformly in n. Small-grade component DAGs and Smith forms are decidable evidence; they cannot certify (22) for the tower.

9. Final boundary

Desk-proved. The “integral ledger saturation” problem split in the wrong place. Full table/price relation lattices are automatically saturated, and the finite transformation monoid completely controls count arithmetic: aperiodic reset structure contributes no torsion, while mixed monoids contribute only periods already present in maximal subgroups.

Desk-proved. These facts close the entire named irreversible family and strictly enlarge the complete tower theory to all $W = \mathrm{id}$ observers and all strongly connected reset observers, integrally.

Wall. What remains is a rational, not integral, local-to-global theorem: fill the signed phase diamonds of a non-strongly-connected observer whose write-below update leaves a boundary trace. Equation (22) is the exact missing lemma. The abstract aperiodic reset diamond proves that transformation-monoid algebra alone cannot establish it; a proof must use the specific reachability and tree geometry of the dyadic machine, or a dyadic instance of that diamond will be the minimal counterexample.

Repository anchors

- AIP-5 §§0–2 and §4: exchange form, group completion, and dirty holonomy.
- `aips/draft/local-classes.md` §2: the observer event language and standard reachable augmented graph.
- `aips/draft/cocycle-classification.md` §§2 and 5: fixed-grade SNF, the open left lift, strong descent, and the earlier ledger sketch corrected here.
- `lean/Adic/Cocycle.lean`: `Exchange` and the subtraction-free public vocabulary.
- `lean/Adic/Machine.lean`: `ActionConfig`, `AddressedOp`, and `actionStep`.